



OPEN Distributed denial-of-service (DDoS) attack detection using supervised machine learning algorithms

S. Abiramasundari[✉] & V. Ramaswamy

Distributed Denial-of-Service (DDoS) attacks have become a critical issue in cyber security. This can lead to a temporary or even prolonged loss of service for users. These attacks mainly target e-commerce platforms, online services, and financial institutions. DDoS attacks need to be detected since they cause serious problems. Supervised machine learning models are effective mechanisms for detecting DDoS attacks. In this paper, a PCA-based Enhanced Distributed DDoS Attack Detection (EDAD) framework has been proposed. Various Machine Learning (ML) algorithms and feature selection techniques have been used to detect DDoS attacks. Support Vector Machine (SVM), Logistic Regression (LR), Random Forest (RF), K-Nearest Neighbours (KNN), Decision Tree (DT) supervised models, and Principle Component Analysis (PCA) feature selection method are used to differentiate between attack and regular traffic. The CICIDS2018, CICIDS2017, and CICDDoS-2019 datasets are used to evaluate the performances of ML algorithms. Various performance metrics of these algorithms are studied and compared to find the best algorithm that yields the highest accuracy. It is found that RF yields the highest accuracy of 98.9% on CICIDS2017. In the CICDDoS2019 dataset, RF and KNN yield a higher accuracy of 98.7. On the CICIDS2018 dataset, SVM gives the highest accuracy of 98.7%.

Keywords DDoS attack, Machine learning, PCA, SVM, Cyberattack

Distributed Denial-of-Service (DDoS) attacks¹ represent one of the most pervasive and disruptive forms of cyber threats in today's digital landscape. DDoS assaults are primarily designed to overload a specific network or online service with an excessive amount of malicious traffic, making it unavailable to authorized users. The word "distributed" refers to the fact that DDoS attacks leverage the strength of several hacked devices or systems spread over various geographic regions, in contrast to classic Denial-of-Service (DoS) attacks, which usually originate from a single source. This distributed nature makes DDoS attacks particularly challenging to mitigate, as they can leverage a vast network of compromised devices, often referred to as a botnet, to orchestrate coordinated assaults on a target. By inundating the target with an excessive volume of requests, data packets, or connection attempts, DDoS attacks exhaust the available bandwidth, server resources, or network infrastructure, thereby disrupting the normal operation of the targeted service and causing downtime.

DDoS attacks employ various techniques and mechanisms to achieve their disruptive objectives. One of technique involves the use of botnets comprised of compromised devices, including computers, IoT devices², and servers, which the attacker remotely controls to carry out coordinated attacks. These botnets enable attackers³ to distribute the attack traffic across multiple sources, making it more challenging to trace and mitigate. DDoS attacks can make use of reflection and amplification techniques, in which the attacker simulates the requests' originating IP address in order to reroute answer traffic in the victim's direction, so increasing the attack's impact and disrupting as much as possible.

The impact of DDoS attacks extends beyond immediate disruptions to the targeted services, encompassing a wide range of adverse consequences for affected individuals, businesses, and society at large. Beyond financial losses incurred due to downtime and service unavailability, DDoS attacks can tarnish the reputation and credibility of organizations, erode customer trust, and result in long-term damage to brand integrity. The motivations driving DDoS attacks vary widely, ranging from ideological reasons, such as hacktivism or political activism, to financial gain through extortion or ransom demands. Understanding the motivations behind DDoS attacks is crucial for developing effective countermeasures and mitigating their impact on targeted entities and the broader digital ecosystem.

SASTRA Deemed to be University, Kumbakonam, Tamil Nadu, India. ✉email: sabiarul07@gmail.com

The major contributions of this paper are listed below.

- Datasets collection.
- Data preprocessing.
- Feature selection using PCA.
- Build classification models using Support Vector Machine (SVM), Logistic Regression (LR), Random Forest (RF), K-Nearest Neighbours (KNN), Decision Tree (DT) algorithms on three datasets.
- Compare the results of all ML algorithms on three datasets.
- Performance evaluation.

Related works

The extensive research on IoT security challenges and the critical need for robust Intrusion Detection Systems (IDS). It highlights the impact of data complexity on IDS accuracy and computational complexity. Recent studies advocate integrating AI techniques, especially Decision Trees, to enhance IDS capabilities. Scholars explore feature selection methods, emphasizing ranking security features for improved efficiency. Evaluation metrics like accuracy, precision, and recall validate IDS performance. Comparative analyses of machine learning methods, including Decision Trees, inform their efficacy. Future research aims to predict cyber-attacks and refine feature selection. Dimensionality reduction techniques are explored to optimize IDS efficiency. Ongoing efforts focus on integrating advanced AI techniques and adapting to evolving cyber threats to enhance IDS⁴.

Research aims to further enhance IDS models by expanding feature selection and data balancing methodologies to predict the types of cyber-attacks. Continuous efforts focus on improving IDS performance, integrating advanced AI techniques, and adapting to evolving cyber threats. These endeavours are crucial in light of the increasing cybercrime landscape and the imperative for organizations to swiftly identify, respond to and recover from security incidents. Theoretical underpinnings of IDS rely on distinguishing between the behaviour of intruders and authorized users. This paper contributes an enhanced intelligent intrusion detection model (E2IDS), incorporating an advanced feature selection approach and data balancing techniques to address real-world challenges. ROC, Precision, F-score, and Recall are among the comprehensive measures used to evaluate the model's performance, which shows that it outperforms current and related work. Future endeavours will involve refining and expanding these methodologies to develop predictive models capable of anticipating various cyber-attack types⁵.

Scholars emphasize the need for further research to validate detection methods across diverse datasets and explore innovative approaches to enhance DDoS defence mechanisms, particularly for small and medium-sized enterprises. Comparative studies have revealed differences in how well various machine learning algorithms identify DDoS assaults, highlighting the importance of selecting the most effective approach based on specific requirements and constraints. Future research directions may involve refining existing detection techniques, incorporating novel features or algorithms, and developing adaptive defence mechanisms capable of mitigating evolving DDoS attack strategies. Additionally, there is a growing interest in real-time detection and response systems that can swiftly identify and mitigate DDoS attacks to minimize their impact on businesses and ensure uninterrupted online operations. Overall, ongoing research efforts are essential to stay ahead of cyber threats and safeguard the integrity and availability of online platforms, especially in the current landscape shaped by the COVID-19 pandemic⁶.

This research in DDoS detection aims to create a quicker and more user-friendly deep learning algorithm for quicker processing and improved results. There's a growing focus on transitioning from unsupervised to supervised learning, including both labelled and unlabelled datasets, to enhance detection capabilities. Exploring non-supervised learning algorithms for DDoS detection, especially with unlabelled datasets, is gaining traction. Comparative analyses with existing research highlight the progress made in the accuracy and performance of detection methods. Adapting DDoS detection mechanisms to effectively address emerging cyber threats is crucial for ensuring robust cybersecurity measures and resilience against evolving attack strategies. Ongoing research efforts are essential for advancing DDoS detection technologies and bolstering cybersecurity against evolving threats⁷.

The integration of deep learning models, such as DNN, CNN, and LSTM, is emphasized in the paper for effective DDoS attack identification in network communications. Studies evaluate these models extensively using datasets like CIC-DDoS2019, employing preprocessing techniques such as feature elimination and normalization to enhance detection performance. Performance metrics like binary and multiclass accuracy assess model efficacy, highlighting the superiority of deep learning-based approaches over baseline methods through comparative analyses. Data preprocessing steps, including duplicate removal and feature selection, are crucial for producing homogeneous datasets, improving training processes, and enhancing detection outcomes. Moreover, normalization techniques like min-max normalization aid in creating balanced datasets across various attack types. Evaluations also demonstrate the real-time processing speed of proposed models, showcasing their efficiency in practical deployment scenarios. These findings underscore the importance of integrating deep learning models and preprocessing techniques to strengthen intrusion detection capabilities and effectively mitigate evolving cyber threats⁸.

This research on DDoS attack detection emphasizes the use of machine learning-based approaches for enhanced security measures. Research has extensively explored various machine learning algorithms, including LSTM, SVM, and logistic regression, for detecting DDoS attacks in network communications. Comparative analyses have highlighted the efficacy of LSTM in achieving high accuracy rates, surpassing traditional methods like SVM and logistic regression. Studies also stress the importance of preprocessing and exploratory data analysis techniques in optimizing model performance. Timing dependencies and sequential patterns in network traffic data are captured by LSTM, which enhances its exceptional performance in detecting malicious activity.

To improve DDoS attack detection in dynamic network environments, future research directions will look at the application of various machine learning techniques and deep learning models in conjunction with real-time monitoring procedures. To provide complete network security solutions, it is also possible to adapt LSTM algorithms to identify different kinds of network assaults and combine them with intrusion detection systems⁹.

The study emphasizes how, despite improvements in security methods, DDoS assaults continue to pose a threat to computer and network infrastructure. Using a public DDoS attack dataset, this study evaluates how well different machine learning techniques classify data. Several well-known classifiers are evaluated, and the best classifier turns out to be Logistic Regression, followed by SVM, KNN, Naïve Bayes, and Stochastic Gradient Classifier. The study does, however, recognize the drawbacks of offline analysis and recommends the necessity of real-time detection techniques to deal with the dynamic nature of DDoS attacks. To improve the detection of DDoS attacks, more studies may look at incorporating real-time monitoring processes¹⁰.

Cyber security and intrusion detection emphasize the increasing importance of efficient detection systems, particularly in light of the proliferation of the Internet of Things (IoT) and intelligent devices. With the rise of real-life applications and the exponential growth of data, constructing robust Intrusion Detection Systems (IDS) has become paramount for preserving data integrity, confidentiality, and availability. Among the various threats, DDoS attacks provide a serious problem, primarily targeting asset availability and necessitating effective IDS implementations. Recent research has focused on leveraging Artificial Intelligence (AI) to develop intelligent IDSs capable of mitigating such threats. This study proposes a novel approach for DDoS attack detection, employing a Decision Tree (DT) algorithm and an enhanced Gini index feature selection method. The analysis, which is based on the UNSW-NB15 dataset, performs better than baseline models that make use of more sophisticated algorithms, with an overall accuracy of 98%. Through efficient reduction of data dimensionality and reduction of false positives, the suggested model demonstrates good recall and accuracy rates for a range of attack types. Additionally, the model shows promise for detecting DDoS attacks in real-world scenarios, especially in settings with limited resources. In summary, this study highlights the effectiveness of utilizing intelligent detection systems to improve network security and efficiently mitigate DDoS attacks¹¹.

Cloud computing deals with various benefits like increased availability and scalability, but it's susceptible to Distributed Denial-of-Service (DDoS) attacks. Early detection of DDoS attacks is challenging, and current methods often detect DDoS at the SDN controller level. A new framework addressed¹² this issue by incorporating multiple techniques, including Recursive Feature Elimination (RFE), Density-Based Spatial Clustering (DBSCAN), and time series techniques. The proposed RDAER model outperformed existing methods on the CICDDoS 2019 dataset, achieving an accuracy level of 99.92% and a fast detection time of 20 s.

This paper analyses AI-based tools and techniques used to analyze, classify, and detect Distributed Denial-of-Service (DDoS) attacks in IoT networks from 2019 to 2023. It provides understandings into the existing research background for designing AI-based models to detect DDoS attacks. The study also explores IoT botnet lifecycle, botnet families, DDoS attack taxonomy, tools used, publicly available datasets, AI techniques, and popular software for ML/DL modelling. The findings highlight research challenges and future directions for developing reliable methods for identifying and categorizing DDoS attacks¹³.

Methodology

This research aims to address the significant threat produced by Distributed Denial of Service (DDoS) attacks in cybersecurity and proposes a framework PCA-Based EDAD for detecting these attacks. In the proposed framework, various supervised machine learning algorithms have been used to detect the attacks. Figure 1 shows the architecture of this framework, which comprises the phases listed below: To detect and evaluate the effectiveness of DDoS attacks and benign scenarios, various techniques such as dataset collection, preprocessing of data, selecting features, splitting data into sets for training and testing, and building a supervised machine learning model known as SVM, Logistic Regression, Random Forest, KNN, and Decision Tree are employed.

Dataset

The University of New Brunswick's Canadian Institute for Cybersecurity (CIC) produced the CICIDS2018^{14,15}, CICIDS2017^{14,15}, and CICDoS-2019^{14,15} databases, which are essential resources for cybersecurity studies. They offer large datasets of labelled and unlabelled network traffic spanning a variety of benign and harmful activities in controlled situations. These datasets cover a variety of attack types, including botnet instances, DDoS attacks, and brute-force attacks. While CICIDS-2018 and CICIDS-2017 provide a general overview of cyber risks, CICDDoS-2019 focuses exclusively on attacks known as Denial-of-Service (DoS). These three datasets are used to train and test the ML algorithms.

Data preprocessing

To increase the quality of datasets, data preparation is essential for machine learning and data analysis. Many important actions were performed on the IDS dataset. To ensure the consistency and completeness of the dataset, missing values are addressed using techniques like imputation or deletion. To ensure that the dataset remained intact, null values—which represent the lack of data—are carefully examined. To avoid computational mistakes, infinite values that resulted from mathematical processes are also handled. To ensure the accuracy of the data, duplicate entries are found and removed. Finally, the process of sorting the dataset allowed for a systematic arrangement that helped in the analysis that followed. Overall, careful preprocessing provided a strong basis for insightful analysis of the IDS dataset.

Label encoding

A basic method for converting categorical data into numerical format in machine learning classification problems is label encoding. Using this approach, a unique integer value is assigned to each separate category

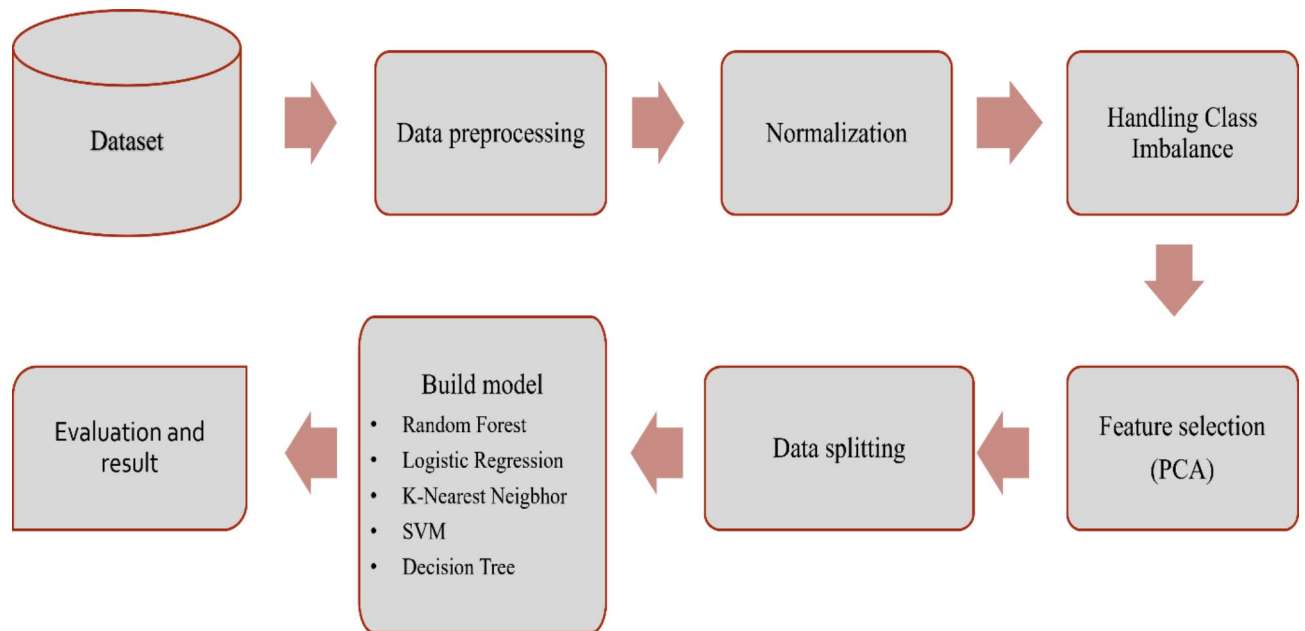


Fig. 1. Proposed model PCA-based EDAD-architecture.

or label within a categorical variable. Label encoding is mostly used in machine learning classification to allow algorithms to operate on categorical data because most algorithms require numerical input. Label encoding enables classifiers to efficiently read and analyse data by expressing category variables as integers. With the Label Encoder, labels can be transformed into numerical representations, making it a useful tool for preparing categorical data in machine learning classification problems.

Normalization

Normalization is a vital preprocessing technique in machine learning classification that aims to standardize the scale of numeric features within a dataset. Its main goal is to rescale each feature to a common scale, usually between 0 and 1 or -1 and 1 , so that all features contribute equally to the learning process. Normalization enhances the convergence speed and stability of the optimization algorithm, facilitating smoother gradient descent during training. StandardScaler is a machine learning normalization approach that helps achieve this by standardizing the scale of numerical characteristics inside a dataset. It creates a standard normal distribution by transforming the data distribution to have a mean of 0 and a standard deviation of 1. This procedure guarantees that every feature contributes equally to the learning process, which is crucial for algorithms that are sensitive to feature sizes. The following formula may be used to standardize a feature with StandardScaler:

$$z = (x - \mu) / \sigma$$

where the feature's standardized value is represented by z , the original value is represented by x , the feature value mean is represented by μ , the feature value's standard deviation is represented by σ .

Handling class imbalance

Class imbalance in machine learning classification occurs when two classes have significantly different numbers of examples in the dataset one class has significantly more instances than the other. Therefore, biased models may predict events from the minority class with less accuracy. Techniques such as oversampling can be used to solve class imbalance. To adjust the class distribution and provide the classifier with a more representative dataset, oversampling involves adding more cases to the minority class. This can be done by creating synthetic samples using techniques like SMOTE. Oversampling helps classifiers better understand the properties of the minority class by increasing its representation, which results in more accurate predictions. To ensure that oversampling effectively lessens class imbalance without leading to biases or issues with overfitting, it is essential to closely evaluate the model's performance. In machine learning classification problems, oversampling is a useful technique for managing class imbalance and improving the accuracy and resilience of models in unbalanced scenarios.

Feature selection

Feature selection is a crucial step in machine learning aimed at identifying and selecting the most relevant and informative features from the dataset while discarding irrelevant or redundant features. Enhancing interpretability, decreasing overfitting, and increasing model performance are the main objectives of feature selection, which focuses on the most discriminative characteristics that make a substantial contribution to the classification job.

One popular method for reducing dimensionality in feature selection for machine learning classification is Principal Component Analysis (PCA). PCA is used to identify principal components, which are linear combinations of features that account for the majority of the variance in the data. The Principle Component Analysis (PCA) method efficiently decreases the dimensionality of a dataset without compromising its information by keeping a selection of main components that account for most of the variation. By converting the original feature space into a lower-dimensional subspace, PCA can be used in classification tasks to reduce noise and redundancy in the data while maintaining its discriminative strength. Classifiers can concentrate on the most crucial elements of the data by employing PCA for feature selection, which enhances model performance and efficiency. To ensure that the chosen features maintain sufficient selective information for precise classification, it is crucial to achieve a balance between dimensionality reduction and information preservation.

Supervised machine learning

Supervised machine learning plays a pivotal role in the detection of Distributed Denial-of-Service (DDoS) attacks, a critical security concern in cyber networks. Supervised learning algorithms provide a logical foundation for differentiating between benign network traffic and malicious activity. These algorithms learn from labelled training data to provide predictions or classifications. Supervised machine learning algorithms can identify patterns that point to DDoS assaults and make judgments about the existence of unusual activity by training on past data that includes both benign and malicious occurrences. The models can recognize abnormalities in behaviour that point to DDoS attacks courtesy of this training, which also sets off the proper defences. To protect the availability and integrity of vital network resources, network defenders are equipped with the supervised learning paradigm, which enables them to proactively identify and neutralize DDoS attacks.

Random forest

Random forest¹⁶ is a popular machine learning technique that is both powerful and versatile regarding classification. As part of the ensemble learning family, it integrates predictions from several different models to obtain a final prediction that is more reliable and accurate. The Random Forest method generates a lot of decision trees during training. Random feature and data sample choices are used to build each decision tree separately from the training dataset. This strategy makes sure that each tree captures distinct characteristics of the underlying data distribution by creating variation among the trees. Random Forest aggregates the predictions from each decision tree separately via a voting mechanism.

SVM

One powerful method for supervised machine learning is the Support Vector Machine (SVM)¹⁷. It splits different classes optimally by locating the best hyperplane in the feature space. The approach optimizes the margin by calculating the distance between the support vectors, or closest data points from each class, and the hyperplane. SVM can handle datasets that are both linearly and non-linearly separable by using a range of kernel functions to transfer the input features into a higher-dimensional space where a hyperplane may successfully separate the classes. SVM solves an optimization issue that maximizes the margin and reduces classification errors to learn the hyperplane's parameters during training. Because of its margin maximization aim, support vector machines (SVM) are particularly good at handling high-dimensional datasets with complicated decision boundaries and are resistant to overfitting.

KNN

An efficient machine learning technique for classification is the K-Nearest Neighbors (KNN)¹⁷ algorithm. The majority class of a new data point's k nearest neighbors in the feature space determines the class label that is assigned to it in KNN. Rather than explicitly training the model, the algorithm takes the training dataset and memorizes it. Using Euclidean distance or other distance metrics, KNN computes the distance between each training instance and the new data point during classification. Then, based on a majority vote among these neighbors, it chooses the k closest neighbors and applies the class label to the newly discovered data point. One of the most important parameters in KNN is the number of neighbors, or k, which influences the model's performance and decision bounds. KNN is easy to learn and use, but depending on the chosen k value and distance measure, it can be computationally expensive for big datasets.

Logistic regression

Logistic regression¹⁷ is a significant machine learning approach mostly used in binary classification. Despite its name, the model that determines the probability that an instance belongs to a particular class is linear. A logistic function also referred to as the sigmoid function, is used in logistic regression to explain how the independent variables and the outcome's probability are related. To minimize the logistic loss function, it adjusts the model's parameters during training, often by utilizing optimization techniques like gradient descent. A threshold value, often 0.5, is used to set the decision boundary, over which occurrences are categorized into one class and below which they are classified into the other. Logistic Regression is a straightforward technique that can efficiently handle both linear and nonlinear correlations between data and target variables.

Decision tree

The Decision Tree¹⁶ algorithm is a versatile machine learning method primarily used for classification. It builds a structure similar to a tree, with each leaf node denoting a class name and each inside node representing a feature property. Using measures like Gini impurity or entropy, the method iteratively divides the feature space according to attribute values during training to optimize information gain or purity. Decision trees follow a route established by the input instance's feature values as they go from the root to the leaf nodes of the tree.

to make judgments. After a succession of if-else conditions based on feature thresholds, the decision process assigns the instance to a certain class label. Decision trees can handle both numerical and categorical data and are interpretable and intuitive. However, they may be prevented from overfitting, particularly with large or complicated datasets by employing ensemble approaches like Random Forests or pruning algorithms. To estimate the performance of our suggested approach, various measures such as Accuracy, Precision, Recall and F1 score have been calculated. These measures are utilized to compare the effectiveness of earlier approaches and are defined as shown below.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$\text{F Score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

FP and FN are the numbers of false positives and negatives. TP and TN are the numbers of true positives and negatives.

Results and discussion

In this research, the significant issue of Distributed Denial-of-Service (DDoS) attacks have been examined, which pose a significant risk to people and companies alike. An approach has been built that differentiate the DDoS attacks and regular network traffics using supervised machine learning.

Three different datasets such as CICIDS2018, CICIDS2017, and CICDoS-2019 have been used to train and test the models. Each dataset has 79 features related to network Traffic. After applying the data preprocessing, the 79 features are reduced to 61 features. Before applying feature selection, the pre-processed data are split. The models have also been improved in terms of performance and interpretability by addressing the issue of class imbalance and ensuring that they have learnt from both majority and minority classes accordingly. 30% has been used to evaluate the model and 70% has been used to train it. Then, PCA has been applied for feature selection on test dataset.

Then, the five machine learning algorithms Random Forest, Logistic regression, K-Nearest Neighbor, Support Vector Machine, and Decision Tree have been used. These algorithms are trained to detect DDoS attacks. The performance of each model for each dataset has been examined to find the best performance among these algorithms that yield the best accuracy.

The effectiveness of the classifier system is evaluated using a confusion matrix. It provides a summary of the predictions made by the model against the actual labels in the dataset. For each dataset, confusion matrix is used to assess the performance of the machine learning models. The confusion matrices of ML models are displayed in Fig. 2.

Other evaluation metrics such as precision and recall are analysed for each model. Precision recall curve is used to represent the visual depiction of the trade-off between accuracy and recall across various categorization thresholds provided by the precision-recall curve. This graphical representation makes it easier to understand how the classifier performs when the class assignment thresholds are changed. The generated precision, recall curves are shown in Fig. 3.

The performance metrics for each dataset have been calculated separately, tabulated and visualized. It shows the accuracy, precision, recall, and f1 score for all machine learning algorithms. In the CICIDS2017 dataset, Random Forest yields higher accuracy than other algorithms. KNN and Decision Tree accuracy are nearly close to the random forest accuracy. RF yields 98.9% accuracy, whereas KNN and decision tree produced 98.7% and 98.4% of accuracy, respectively. Table 1 and Fig. 4 shows the result of the CICIDS2017 dataset.

In the CICIDS2018 dataset, SVM and KNN yield higher accuracy rates of 98.7% and 98.6%, respectively, than other algorithms. Table 2 and Fig. 5 shows the comparison of the CICIDS2018 dataset.

In the CICDDoS2019 dataset, Random Forest and KNN both yield a higher accuracy of 98.7%. Decision Tree accuracy (98.1%) is close to the Random Forest and KNN accuracy. Table 3 and Fig. 6 shows the result of the CICDDoS2019 dataset.

The overall accuracy comparison of each dataset is represented in Fig. 7.

Conclusion

This study focuses on countering Distributed Denial-of-Service (DDoS) attacks, which pose a serious risk to people and businesses. A PCA-based EDAD framework has been created to differentiate DDoS attacks from normal network data using supervised machine learning. Three datasets with network traffic-related features, CICIDS2018, CICIDS2017, and CICDoS-2019, have been used for training and testing the model. Machine learning methods such as Random Forest, Logistic Regression, K-Nearest Neighbor, Support Vector Machine, and Decision Tree are used after Data Preprocessing, Feature Selection, and Class Imbalance correction. Every Algorithm's performance are assessed using every dataset. Random Forest produced the best accuracy (98.9%) in CICIDS2017, followed closely by KNN and Decision Tree. SVM and KNN achieved the maximum accuracy 98.7% and 98.6%, respectively in CICIDS2018. KNN and Random Forest both produced the greatest accuracy in CICDDoS2019 (98.7%).

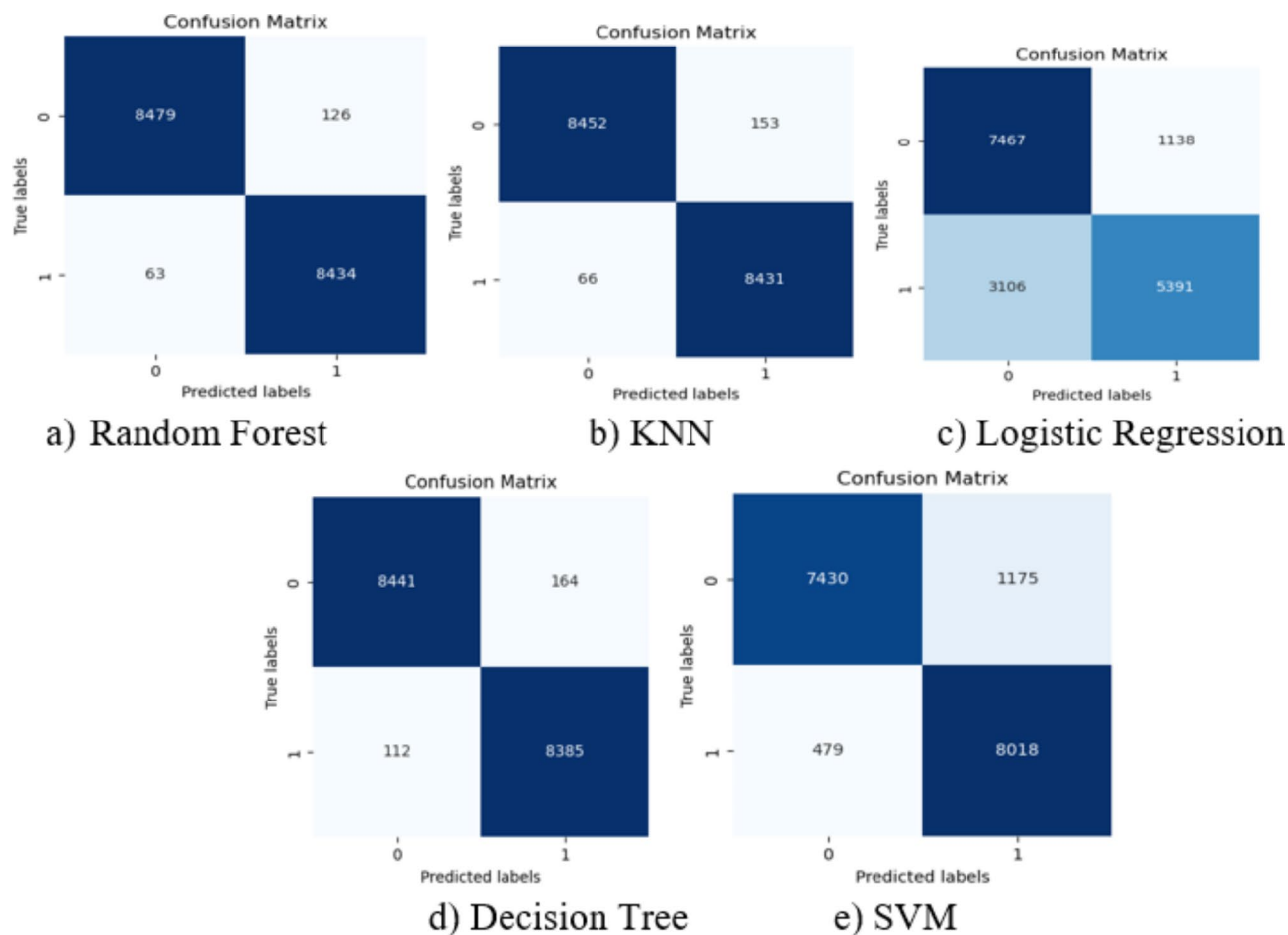


Fig. 2. Confusion matrices of ML algorithms on the test dataset.

In future, it may be possible to identify ensemble techniques that combine the benefits of many algorithms to improve the performances., feature engineering techniques may be found to enhance the model's interpretability and performance, particularly when high-dimensional data are included. Network breaches may be identified with detailed patterns using deep learning techniques like convolutional neural networks (CNN) and Recurrent Neural Network (RNN).

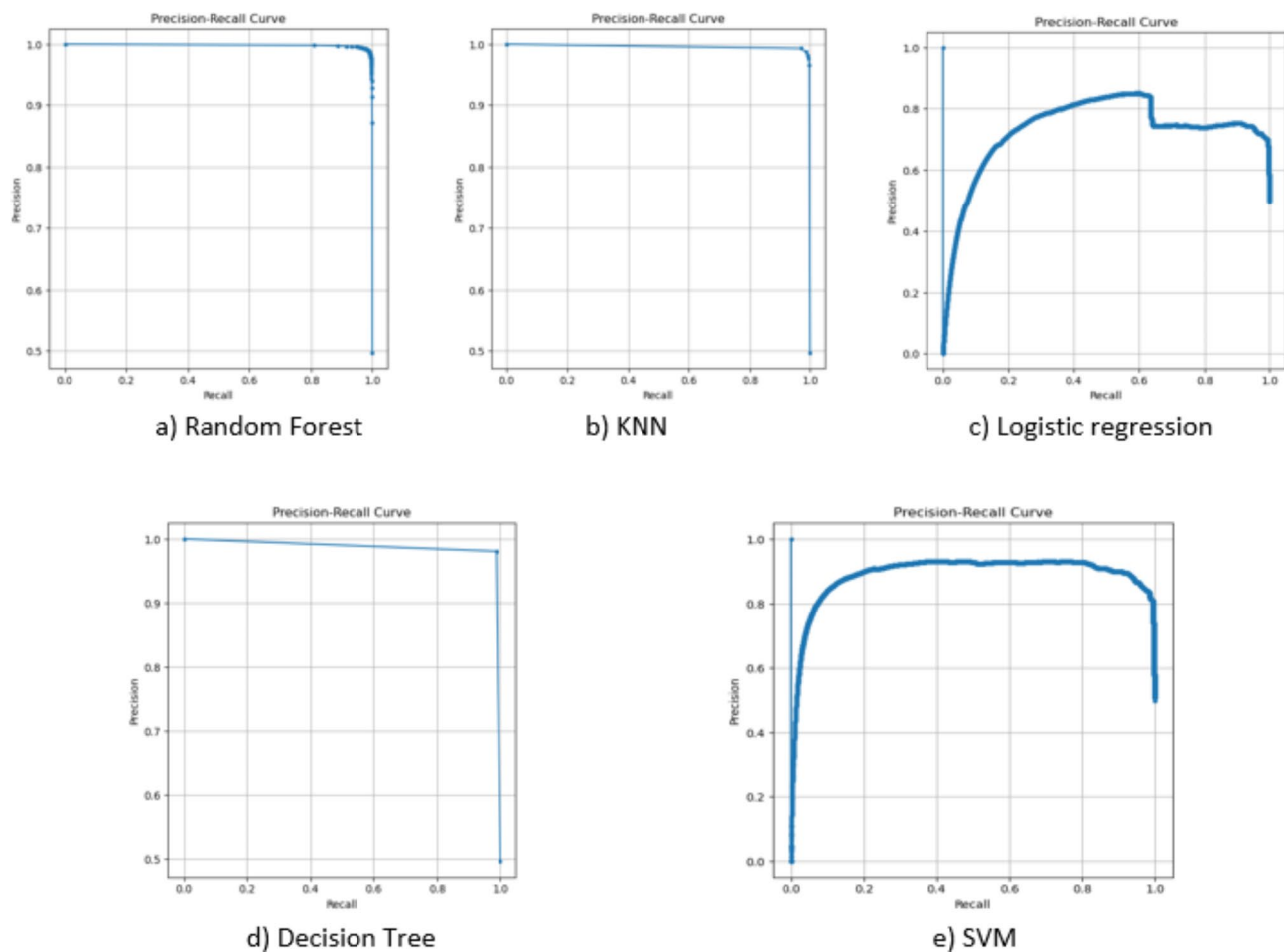


Fig. 3. Precision-recall curves of ML algorithms on test dataset.

Classifiers	Precision	Recall	F1-score	Accuracy
Random forest	0.98	0.99	0.98	0.98
KNN	0.98	0.99	0.98	0.98
Logistic regression	0.82	0.63	0.71	0.75
Decision tree	0.98	0.98	0.98	0.98
SVM	0.87	0.94	0.94	0.9

Table 1. Performance metrics of ML algorithms on CICIDS2017 dataset.

CICIDS-2017

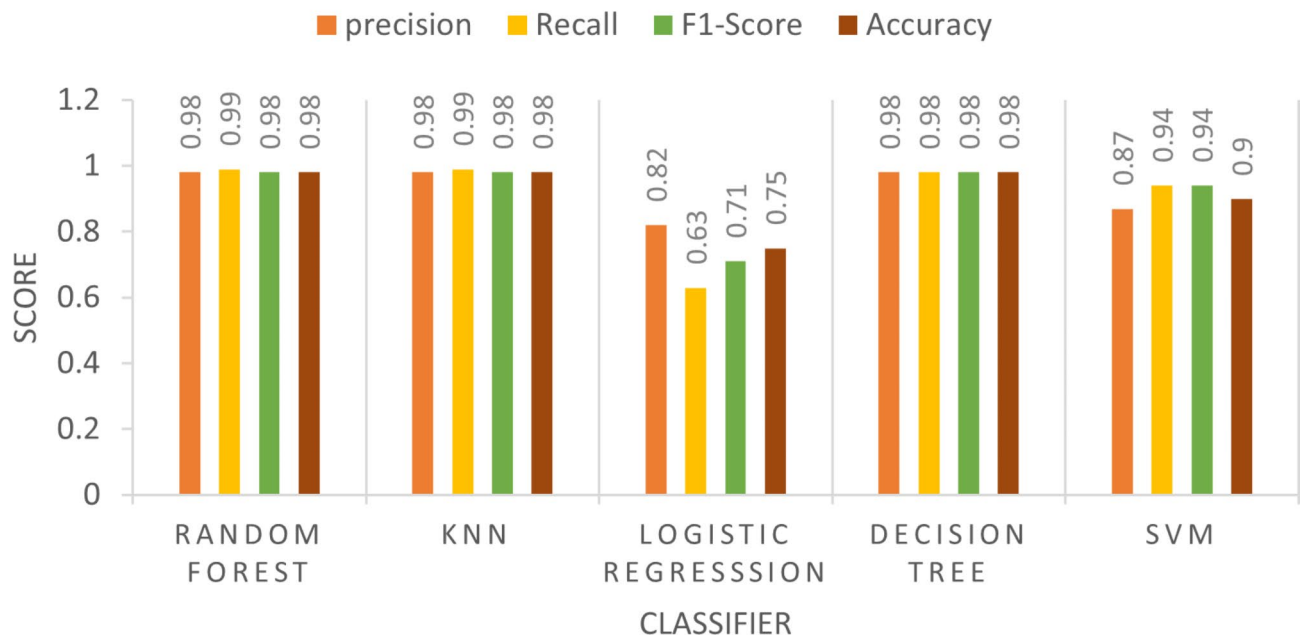


Fig. 4. Scores comparison of the models applied in CICIDS2017.

Classifiers	Precision	Recall	F1-score	Accuracy
Random forest	0.95	0.96	0.96	0.96
KNN	0.99	0.96	0.97	0.97
Logistic regression	0.99	0.95	0.97	0.97
Decision tree	0.95	0.96	0.96	0.96
SVM	0.99	0.95	0.97	0.97

Table 2. Performance metrics of ML algorithms on CICIDS2018 dataset.

CICIDS2018

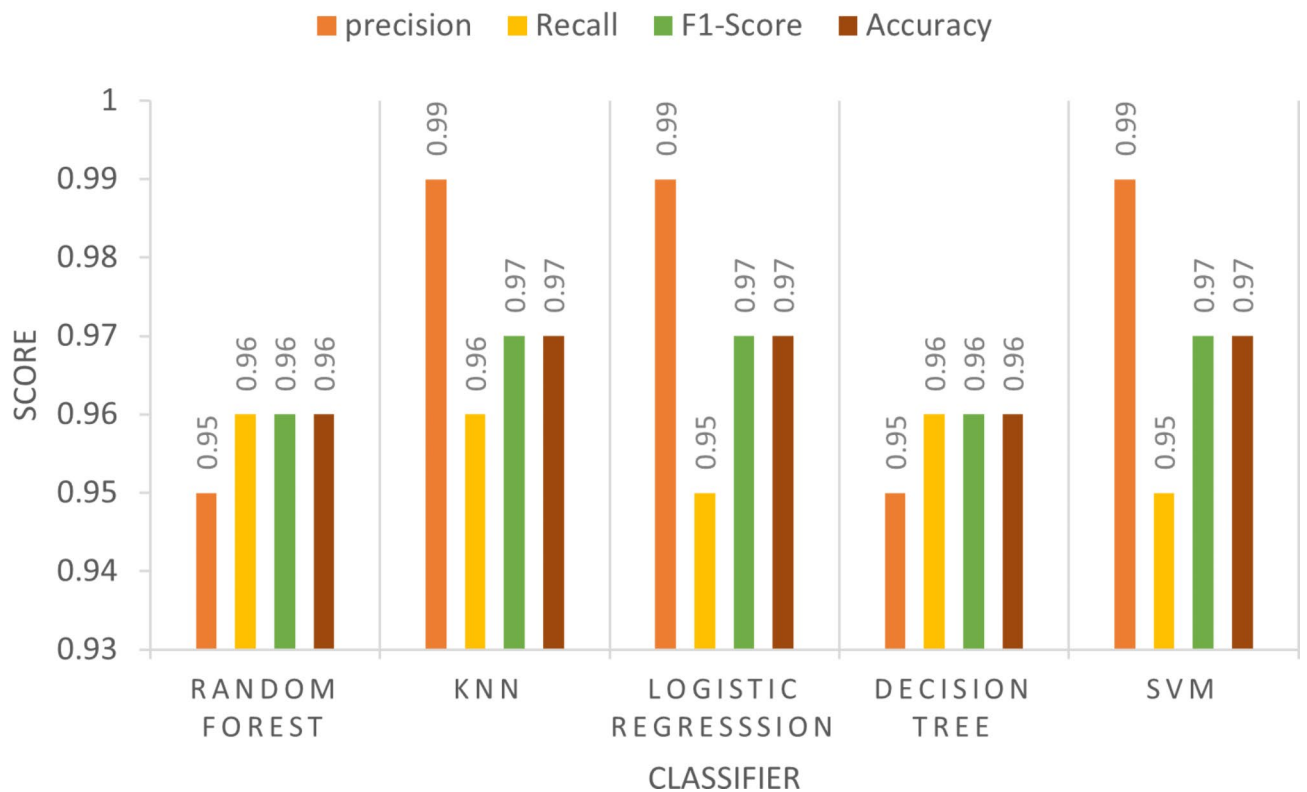


Fig. 5. Accuracy comparison of the models applied in CICIDS2018.

Classifiers	Precision	Recall	F1-Score	Accuracy
Random forest	0.98	0.98	0.98	0.98
KNN	0.98	0.98	0.98	0.98
Logistic regression	0.84	0.9	0.87	0.86
Decision tree	0.98	0.98	0.98	0.98
SVM	0.95	0.97	0.96	0.96

Table 3. Performance metrics of ML algorithms on CICDDoS2019 dataset.

CICDDOS-2019

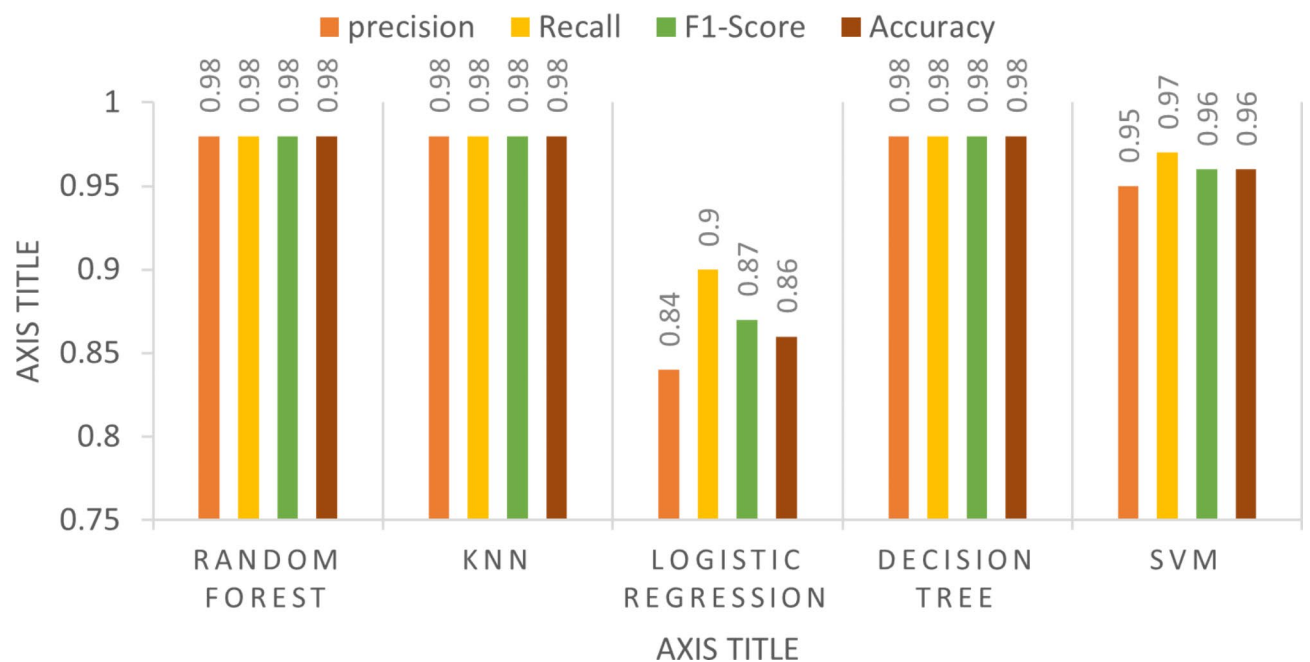


Fig. 6. Accuracy comparison of the models applied on CICDDoS2019.

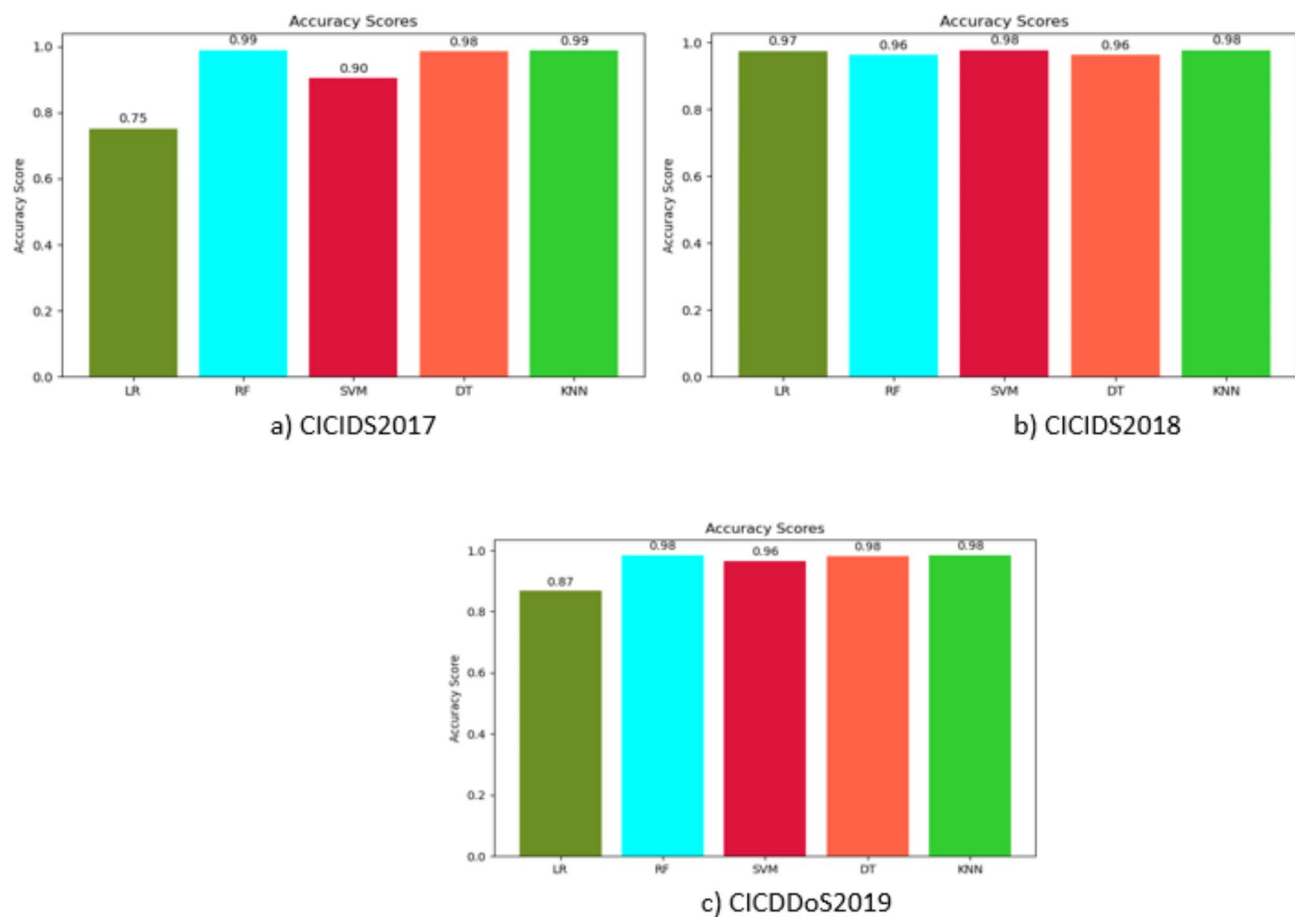


Fig. 7. Overall comparison of performances of ML algorithms on three datasets.

Data availability

The link for accessing the datasets is also given in the manuscript itself. The following is the link to access datasets. <https://drive.google.com/drive/u/0/folders/1E8a1J6aMbmOhu61epOUYBIPF-KBthDp3>.

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Declarations

Competing interests

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Research involving human participants and/or animals

All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional research committee.

Additional information

Correspondence and requests for materials should be addressed to S.A.

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